The angular momentum of the rod and two balls about the pivot is

The angular-momentum law gives

By the definition of angular acceleration ,

Equations give

While both are on the rod, they circle the pivot. Ball has the following tangential and normal accelerations:

Let be the force normal to the rod on ball , and the force on ball along the rod. Newton's second law gives

Equations give

For ball , the force normal to the rod on ball has magnitude ; the force along the rod on ball has magnitude . The same analysis gives

By , balls , have equal static-friction limits; show that ball needs more friction than ball . During rotation, ball reaches the limit first, so ball slides first. As ball starts to slide, let the angle